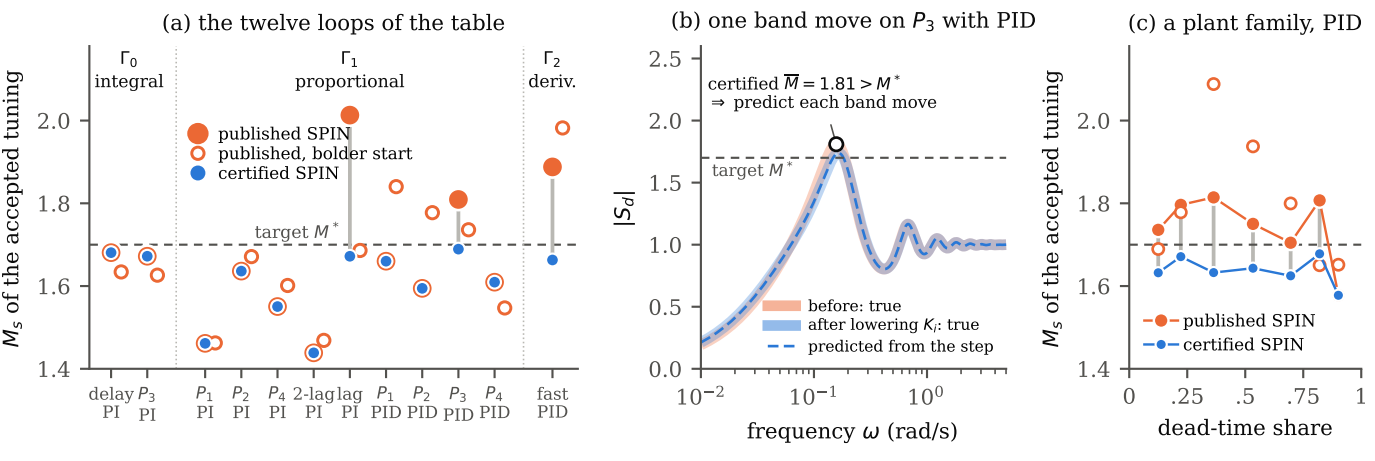
(a) the twelve loops of the table
Γ0 integral
Γ1 proportional
Γ2 deriv.
published SPIN
published, bolder start
certified SPIN
target M*
Ms of the accepted tuning
2.0
1.8
1.6
1.4
delay PI
P3 PI
P1 PI
P2 PI
P4 PI
2-lag PI
lag PI
P1 PID
P2 PID
P3 PID
P4 PID
fast PID
(b) one band move on P3 with PID
certified M = 1.81 > M*
⇒ predict each band move
target M*
|Sd|
2.5
2.0
1.5
1.0
0.5
0.0
before: true
after lowering Ki: true
predicted from the step
10−2
10−1
100
frequency ω (rad/s)
(c) a plant family, PID
Ms of the accepted tuning
2.0
1.8
1.6
1.4
published SPIN
certified SPIN
0
.25
.5
.75
1
dead-time share